

Wei Wang (王尉)

Ph.D. Candidate in Mathematics, Peking University
School of Mathematical Sciences
wwmath166@outlook.com

EDUCATION

- **Peking University, School of Mathematical Sciences** Sept. 2022 – Present
Ph.D. Candidate in Mathematics; Advisor: Prof. Zhifei Zhang
Beijing, China
- **Zhejiang University, School of Mathematical Sciences** Sept. 2018 – June 2022
B.S. in Mathematics and Applied Mathematics; Advisor: Prof. Ting Zhang
Hangzhou, China
Thesis: *Homogenization Problem of Second Order Elliptic Systems with Lower Order Terms*

RESEARCH INTERESTS

- **Research areas:** Geometric variational problems; elliptic and parabolic partial differential equations.
- **Selected topics:** Homogenization theory; semilinear elliptic and parabolic equations; harmonic maps and their flows; liquid crystal theory; nonlocal operators.

PUBLICATIONS AND ACCEPTED PAPERS

6. **Uniform estimates of Landau–de Gennes minimizers in the vanishing elasticity limit with line defects.**
Accepted for publication in *SIAM Journal on Mathematical Analysis*.
arXiv: [2508.01811](https://arxiv.org/abs/2508.01811)
Coauthors: Haotong Fu and Huaijie Wang.
5. **Stratification and rectifiability of harmonic map flows via tangent measures.**
Mathematische Annalen **395** (2026), Article No. 92.
DOI: [10.1007/s00208-026-03524-3](https://doi.org/10.1007/s00208-026-03524-3)
Coauthors: Haotong Fu, Ke Wu, and Zhifei Zhang.
4. **Landau–de Gennes model with sextic potentials: asymptotic behavior of minimizers.**
Accepted for publication in *Memoirs of the European Mathematical Society*.
arXiv: [2404.00677](https://arxiv.org/abs/2404.00677)
Coauthor: Zhifei Zhang.
3. **On the sharp constants in the regional fractional Sobolev inequalities.**
SN Partial Differential Equations and Applications **6** (2025), Article No. 15.
DOI: [10.1007/s42985-025-00317-2](https://doi.org/10.1007/s42985-025-00317-2)
Coauthors: Rupert L. Frank and Tianling Jin.
2. **Uniform estimates of resolvents in the homogenization theory of elliptic systems.**
Journal of Differential Equations **370** (2023), 1–65.
DOI: [10.1016/j.jde.2023.06.015](https://doi.org/10.1016/j.jde.2023.06.015)
Sole author.
1. **Homogenization theory of elliptic systems with lower order terms in dimension two.**
Communications on Pure and Applied Analysis **22** (2023), no. 3, 787–824.
DOI: [10.3934/cpaa.2023010](https://doi.org/10.3934/cpaa.2023010)
Coauthor: Ting Zhang.

PREPRINTS AND MANUSCRIPTS

8. **A note on an approximate fixed point theorem for VMO maps.**
Preprint.
arXiv: [2606.15670](#)
Coauthor: Haotong Fu.
7. **Sharp convergence rates for parabolic Green's functions in time-independent periodic homogenization.**
Preprint.
arXiv: [2606.12101](#)
Sole author.
6. **Asymptotics of minimizers for Ginzburg–Landau-type functionals in high dimensions.**
Preprint.
arXiv: [2605.04442](#)
Coauthors: Giacomo Canevari and Haotong Fu.
5. **Smooth convergence of Landau–de Gennes minimizers in the Lyuksyutov regime.**
Manuscript.
Coauthors: Haotong Fu and Huaijie Wang.
4. **Improved convergence of Landau–de Gennes minimizers in the vanishing elasticity limit.**
Preprint.
arXiv: [2507.14955](#)
Coauthors: Haotong Fu and Huaijie Wang.
3. **Existence of weak solutions for two-phase matrix-valued harmonic map flows.**
Preprint.
arXiv: [2506.16278](#)
Coauthors: Wei Wang (王伟) and Zhifei Zhang. The coauthor Wei Wang (王伟) is a different person.
2. **Fine structure of rupture set for semilinear elliptic equations with singular nonlinearity.**
Preprint.
arXiv: [2411.16048](#)
Coauthor: Zhifei Zhang.
1. **Quantitative stratification and sharp regularity estimates for supercritical semilinear elliptic equations.**
Preprint.
arXiv: [2408.06726](#)
Coauthors: Haotong Fu and Zhifei Zhang.

RESEARCH VISITS AND PROGRAMS

- **University of Verona, Department of Computer Science** Mar. – Apr. 2026
Visiting Ph.D. Student; Host: Prof. Giacomo Canevari Verona, Italy
- **HKUST, Department of Mathematics** June – Sept. 2021
Summer Research Program; Advisor: Prof. Tianling Jin Hong Kong, China

SELECTED TALKS

- **Workshop on Fluid Dynamics Equations, Peking University** Jan. 2026

Title: Landau–de Gennes Model with Sextic Potentials: Asymptotic Behavior of Minimizers

- **Jinhua Conference on Harmonic Analysis, Zhejiang Normal University** Nov. 2025
Title: Stratification for Harmonic Map Flows via Tangent Measures: Rectifiability and Regularity
- **3rd National Forum for Doctoral Students in Frontier Mathematics, Peking University** Nov. 2025
Title: Vanishing Elasticity Limit and Defect Analysis in the Landau–de Gennes Model
- **Conference on Theory and Computation of Quasicrystals, Nankai University** Aug. 2025
Title: Improved Convergence of Landau–de Gennes Minimizers in the Vanishing Elasticity Limit
- **PDE and Geometry Seminar, Zhejiang University** June 2025
Title: Rectifiability of the Stratification for Harmonic Map Flows via Tangent Measures

GRANTS AND PROJECTS

- **Physical Modelling and Mathematical Theory of Liquid Crystals,** 2023 – 2028
Participant, NSFC, Grant No. 12288101.
- **Physical Modeling and Mathematical Theory of Liquid Crystal Systems,** 2023 – 2028
Participant, National Key R&D Program of China, Grant No. 2023YFA1008801.

TEACHING EXPERIENCE

- **Peking University, Teaching Assistant**
 - Advanced Mathematics (A, B, and C), various semesters. 2022 – 2025
 - Real Analysis; Functional Analysis (I); Harmonic Analysis. 2023 – 2025
- **Zhejiang University, Teaching Assistant**
 - Mathematical Analysis (I and II). 2021 – 2022

HONORS AND AWARDS

- **First Prize**, Graduate Research Exhibition, School of Mathematical Sciences, Peking University. Apr. 2025
- **Third Prize**, Graduate Research Exhibition, School of Mathematical Sciences, Peking University. Apr. 2024
- Elite Program for Graduate Students in Mathematics, Peking University. Oct. 2023
- Nomination Award, ICCM Creative Undergraduate Thesis. Aug. 2022
- **First Prize**, 11th Chinese Mathematics Competitions Finals. Apr. 2021
- Nomination Award, S.-T. Yau College Student Mathematics Contest. June 2021

TECHNICAL SKILLS

- **Software:** $\text{\LaTeX}$, MATLAB, Mathematica, Python, Lean.
- **Languages:** Mandarin Chinese (native), English (fluent).